Do not write outside the box

0 4 Group 7 elements are known as the halogens.

All atoms of Group 7 elements contain protons, neutrons and electrons.

0 4 . 1 What is the order of discovery of the proton, neutron and electron?

[1 mark]

Tick (✓) **one** box.

electron → neutron → proton ☐

electron → proton → neutron ☐

neutron → proton → electron ☐

proton → electron → neutron ☐

0 4 . 2 **Table 1** shows the mass of a proton and of an electron.

Table 1

Name of particle	Mass in kg
Proton	1.673×10^{-27}
Electron	9.109×10^{-31}

Calculate how many times heavier a proton is than an electron.

[2 marks]

_____

_____

_____

_____

Times heavier a proton is than an electron = _____

Do not write outside the box

A bromine atom can be represented as $^{81}_{35}Br$.

0 4 . 3 What is the number of neutrons in this bromine atom?

[1 mark]

_____

0 4 . 4 What is the number of electrons in a bromide **ion**?

[1 mark]

_____

0 4 . 5 Chlorine has two isotopes.

Table 2 shows the percentage abundance of the two isotopes of chlorine.

Table 2

Isotope	Percentage (%) abundance
$^{35}_{17}Cl$	75.77
$^{37}_{17}Cl$	24.23

Calculate the relative atomic mass (A_r) of chlorine.

Give your answer to 2 decimal places.

[3 marks]

_____

_____

_____

_____

_____

Relative atomic mass (2 decimal places) = _____

Turn over ►

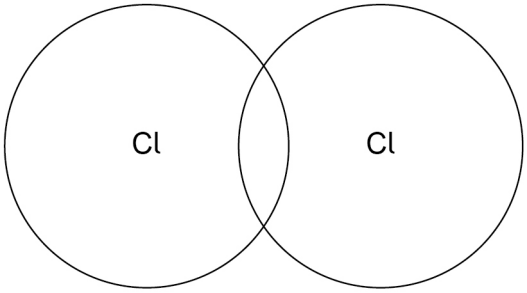
Do not write outside the box

0 4 . 6 **Figure 4** shows the outer shells in one molecule of chlorine (Cl_2).

Complete the dot and cross diagram to show the electrons in the outer shells.

[2 marks]

Figure 4



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